

# The Benefits of Rust and How it Scores Over C



Rust is a modern systems programming language that primarily aims at using code that is memory safe. It has a lot of new features that are not available in the C programming language. Rust makes dangling pointers impossible. It also tests the code examples mentioned in the documentation and therefore keeps the code syntax updated.

**R**ust is a systems programming language that is designed for writing systems software compiled with the self-hosted compiler known as *rustc*, with the low level virtual machine (LLVM) as its backend. The compiler is written in the Rust programming language and is developed by the Mozilla community. It ensures memory safety by setting the invariant in such a way that the creation of dangling pointers is not possible. It is also type safe, that is, it prevents type errors. Rust is a multi-paradigm language that supports procedural, functional and object-oriented styles as well as concurrent computations. It is designed in such a way that it can be used to create large client and server programs that run over the Internet.

## Memory safety

In software development, memory safety is considered to be an important factor as it prevents software errors that can cause security vulnerabilities through RAM accesses. Data race and dangling pointers are some of the memory errors that are common

to most of the languages. Dangling pointers do not occur in Rust because of the special types of pointers used in it, called smart pointers. Memory allocated to smart pointers is freed only when they go out of scope, hence avoiding the formation of dangling pointers, which point to a de-allocated memory.

The data in Rust is immutable, by default, and can be made mutable by adding the keyword *mut*. Rust does not allow the sharing of mutable data between threads; hence, it eliminates race conditions for in-memory data. Instead of sharing, data is copied or owned pointers are moved between tasks. In addition to this, Rust ensures that data is initialised before it is read from anywhere in the program and also does 'bound checking'.

## Installation in Linux and Mac

You can use the *rustup* script to install Rust in Linux and Mac, as follows:

```
$ curl -L https://static.rust-lang.org/rustup.sh -O
```